

# Wito Malik

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## Education

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| Feb 2021 – Jan 2023 | <b>National Taiwan University of Science and Technology (NTUST) – R.O.C.</b><br>Thesis: Virtual testing systems with trajectory-guided control learned by others<br><i>MSc in Electronic and Computer Engineering</i> |
| Jun 2016 – Aug 2020 | <b>Universitas Indonesia (UI) – Indonesia</b><br>Thesis: Development of Medical Image Encryption System Using Byte-Level Base-64 Encoding and AES Encryption Method<br><i>BE in Computer Engineering</i>              |

## Work Experiences

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| Jun 2024 – Now      | <b>Groundhog Technologies, Inc. – R.O.C.</b><br><i>AI Engineer</i> <ul style="list-style-type: none"><li>• Architected an agentic system using Google ADK with fuzzy-matching tool selection algorithms, optimizing tool accuracy and decreasing agent invocation time by 50%.</li><li>• Built a Text-to-SQL agent and integrated Retrieval-Augmented Generation (RAG) via a custom Model Control Protocol (MCP) to standardize backend model communication.</li><li>• Deployed lightweight, privacy-preserving local language models using a Llama.cpp server distributed across Mac Studio and NVIDIA DGX hardware infrastructures.</li><li>• Automated data-driven analysis for telecom site planning and led the system-level integration of modular Open RAN (O-RAN) network applications.</li></ul> |
| Apr 2023 – Jun 2024 | <b>Groundhog Technologies, Inc. – R.O.C.</b><br><i>Software Engineer</i> <ul style="list-style-type: none"><li>• Developed a scalable Flask automation framework to manage data workflows, integrating WordPress and Redmine to streamline task distribution.</li><li>• Engineered large-scale data aggregation architectures using advanced ClickHouse queries, delivering low-latency structured analytics for Tableau and Grafana dashboards.</li></ul>                                                                                                                                                                                                                                                                                                                                                |
| Jan 2022 – Jan 2023 | <b>NTUST and DRC – R.O.C.</b><br><i>Autonomous Systems Research Assistant (UR Project)</i> <ul style="list-style-type: none"><li>• Developed a CARLA simulation testing system covering data gathering to performance assessment, utilizing Open Scenarios and highly detailed 3D maps.</li><li>• Innovated an autonomous vehicle model integrating a control framework from ROACH, combined with a multi-sensor system to elevate industry safety mapping capabilities.</li></ul>                                                                                                                                                                                                                                                                                                                        |
| Feb 2021 – Jan 2022 | <b>NTUST and ITRI – R.O.C.</b><br><i>Computer Vision Research Assistant</i> <ul style="list-style-type: none"><li>• Optimized PyTorch training workflows by implementing complex data augmentation pipelines and deep domain adaptation on LAV models, significantly increasing detection accuracy.</li><li>• Evaluated and replicated state-of-the-art ML architectures from Detectron-based papers to establish standardized internal baselines.</li></ul>                                                                                                                                                                                                                                                                                                                                              |

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| Feb 2019 – Aug 2019 | <b>NTUST – R.O.C.</b><br><i>Software Engineering Intern</i> <ul style="list-style-type: none"><li>• Developed WebSocket solutions for real-time UI-server communication, elevating user engagement and reducing system latency.</li><li>• Designed a smart user access scheduler utilizing advanced scheduling algorithms to continuously optimize server resources.</li><li>• Built and deployed robust microservices on Docker using C#, ensuring high back-end reliability and peak system performance. <a href="https://hackmd.io/@witolmalik/ByPQ1YFeB">https://hackmd.io/@witolmalik/ByPQ1YFeB</a></li></ul> |
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Technical Skills

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| AI & ML          | Google ADK, PyTorch, Detectron, Ollama, Llama.cpp, LangGraph, RAG, ReAct Agents, MCP, Model Benchmarking |
| Programming      | Python, C++ , Java, C#, JavaScript/TypeScript, PHP, Bash/Shell                                           |
| Backend & Web    | Flask, FastAPI, Express.js, Next.js, Angular                                                             |
| Data Engineering | ClickHouse, PostgreSQL, MySQL, MongoDB, Kafka, Schema Registry                                           |
| DevOps & Cloud   | Docker, Kubernetes, Linux, Git, CI/CD, Helm                                                              |
| Simulation       | CARLA, ScenarioRunner, Leaderboard, Multi-Sensor Systems                                                 |

Awards

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- NTUST Scholarship Awardee for MSc (2021-2023)
- Finalist of Indonesian Student Business Competition (2019)
- 3rd Winner of National Annual Informatics Competition: Anforcom (2015)

Languages

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- **Bahasa** Native Speaker
- **English** Full Professional Proficiency
- **Chinese** Capable of Basic

Publication

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1. Malik, M. W. *et al.* Development of Medical Image Encryption System Using Byte-Level Base-64 Encoding and AES Encryption Method in *Proceedings of the 6th International Conference on Communication and Information Processing* (2021), 153–158.